MTH 336(A) Dr. James Ashe
Final Exam Review Monday 25th November, 2013

The final date/time: Wednesday 4th December, 2013 from 10am–12pm.

- Let $S = \mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m$ be a set of m vectors in $\mathbb{R}^n$.
  - Span$S$ is a subspace of $\mathbb{R}^n$.
    - $\dim(\text{Span}S) \leq m$. $\dim(\text{Span}S)$ (the *dimension* of the span of S) is equal to the number of independent vectors in S.
    - Note: Span$S \neq S$, S is not necessarily a subspace (and usually is not).
  - If $m < n$ then Span$S \neq \mathbb{R}^{\text{n}}$ (too few vectors).
  - If $m > n$ then S is linearly dependent (more vectors than necessary).
  - If S is linearly independent and $m = n$ then S is a *basis* for $\mathbb{R}^n$.

- Let R be the *reduced row echelon form* of a $n \times n$ matrix A.
  - If R (or A) has a row of zeros then A is singular.
  - If R has a row of zeros then the system of linear equations $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions or no solution (A is singular).

    ex. $\left[\begin{array}{ccc|c} 1 & 0 & 4 & 5 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right]$ and $\left[\begin{array}{ccc|c} 1 & 0 & 4 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$ have infinitely many solutions.

    ex. $\left[\begin{array}{ccc|c} 1 & 0 & 4 & 5 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 2 \end{array}\right]$ has no solution.
  - If R reduces to the identity then $A\mathbf{x} = \mathbf{b}$ has a unique solution (A is nonsingular).

    ex. $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right]$ and $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$ have unique solutions (a unique solution to $A\mathbf{x} = \mathbf{0}$ is called a *trivial* solution).

- Let A be an $n \times n$ matrix then A is *nonsingular*

$\iff$ A reduces to the identity matrix
$\iff$ A^{-1} exists (A is invertible)
$\iff$ $A\mathbf{x} = \mathbf{b}$ has a unique solution.
$\iff$ $A\mathbf{x} = \mathbf{0}$ has a *trivial solution* ($\mathbf{x} = \mathbf{0}$).
$\iff$ $\det A \neq 0$

- Let A be an $n \times n$ matrix then A is *singular*

$\iff$ A reduces to a matrix with a row of zeros.
$\iff$ A^{-1} does not exists.
$\iff$ $A\mathbf{x} = \mathbf{b}$ has no or infinite solutions.
$\iff$ $A\mathbf{x} = \mathbf{0}$ has a *non-trivial solution* ($\mathbf{x} \neq \mathbf{0}$).
$\iff$ $\det A = 0$

- Let A and B be $n \times n$ matrices then
  - $\det(AB) = \det(A)\det(B)$
  - $\dfrac{1}{\det A} = \det(A^{-1}) \iff \det A \neq 0$ (A^{-1} is invertible)
  - for an A' obtained from A, $\det A = \lambda \det(A')$ for some nonzero $\lambda \in \mathbb{R}$ where the value of λ can be found by using the properties of determinants.
  - $\det A = \det B$ if A and B are similar.

- Eigenvalues and Eigenvectors.
  - $p_A(t) = \det(A - \lambda I_n) = 0 \iff \lambda$ is an eigenvalue.
  - A and B are *similar matrices* if $A = P^{-1}BP$.
    - $\Rightarrow \det A = \det B$.
    - $\Rightarrow p_A(t) = p_A(t)$, i.e., similar matrices have the same eigenvalues.
  - 0 is an eigenvalue of $A \iff A$ is singular.
  - If A has distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$ with corresponding eigenvectors $S = \mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$, then S is linearly independent and A is diagonalizable (this is a sufficient but not necessary condition).
- Linear transformations, $T\colon \mathbb{R}^{\mathrm{m}} \to \mathbb{R}^{\mathrm{n}}$.
  - If T is a linear transformation then there exists a *standard* $m \times n$ *matrix* A such that $\mu_A\colon \mathbb{R}^{\mathrm{m}} \to \mathbb{R}^{\mathrm{n}} = T$.
  - $T[\mathbf{x}] = \mathbf{y}$ is a function given by $A\mathbf{x} = \mathbf{y}$ where $\mathbf{x} \in \mathbb{R}^{\mathrm{m}}$ and $\mathbf{y} \in \mathbb{R}^{\mathrm{n}}$.
  - The i^{th} column of a standard matrix will be $T\left[\mathbf{x}_i\right]$ where $\mathbf{x}_i$ is the i^{th}vector in the standard basis for $\mathbb{R}^{\mathrm{m}}$.
- Be able to compute the following:
  - $\det A$ using cofactors and properties of matrices.
  - A^{-1}.
    - If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $\det A \neq 0$ then $A^{-1} = \begin{bmatrix} \dfrac{d}{ad-bc} & \dfrac{-b}{ad-bc} \\ \dfrac{-c}{ad-bc} & \dfrac{a}{ad-bc} \end{bmatrix}$.
    - If $\det A \neq 0$ and $\det B \neq 0$ then $\left(\begin{bmatrix} A & \mathbf{0} \\ \mathbf{0} & B \end{bmatrix}\right)^{-1} = \begin{bmatrix} A^{-1} & \mathbf{0} \\ \mathbf{0} & B^{-1} \end{bmatrix}$ (see ex 12 in section 2.3)
  - $p_A(t)$.
  - the reduced row echelon form of A.
  - the eigenvalues and eigenvectors for a given matrix.